# Temporally Consistent Exemplar-Based Video Colorization Using Distribution-Aware Perceptual Loss and Time-Distance Weighted Bidirectional Frame Fusion


## Abstract

Exemplar-based video colorization transfers chromatic information from one or more color reference frames to a grayscale video sequence. Although recent deep-learning-based methods have substantially improved the quality of individual colorized frames, temporal inconsistency remains a major limitation. Independently generated or abruptly reference-switched frames can exhibit flickering, blinking, semantic color drift, color bleeding, and sudden transitions between neighboring frames. This paper proposes a temporally consistent exemplar-based video colorization framework integrating two complementary mechanisms: a distribution-aware perceptual loss and a time-distance weighted bidirectional frame-fusion strategy.

First, an advanced perceptual objective is introduced to improve the semantic correspondence between estimated color frames and ground-truth images. Instead of depending exclusively on the Euclidean distance between deep feature tensors, the proposed spatial perceptual loss compares normalized VGG19 feature distributions through a Bhattacharyya-distance formulation. The motivation is to provide a distribution-sensitive penalty capable of distinguishing activation patterns that may have similar aggregate Euclidean errors but different semantic distributions. A temporal perceptual consistency term is further introduced to constrain changes in deep representations between adjacent colorized frames.

Second, a bidirectional between-frame colorization strategy is proposed. Every target frame between two neighboring key frames is colorized twice, once using the left reference and once using the right reference. The two predictions are fused according to normalized temporal distances. Consequently, the influence of one reference decreases progressively while that of the other increases progressively, eliminating the abrupt reference switch inherent in conventional half-interval schemes.

Because full-scale benchmark execution was not available in the current development environment, no unmeasured numerical values are fabricated in this manuscript. Instead, the experimental section presents a reproducible evaluation protocol, mathematically demonstrable properties of the proposed fusion rule, component-wise analytical comparisons, qualitative expected behaviors, ablation hypotheses, and explicit result tables to be completed when benchmark execution becomes available. The analysis shows that the fusion mechanism guarantees continuous convex interpolation across each key-frame interval, whereas the combined perceptual formulation is specifically designed to improve semantic feature discrimination and inter-frame consistency. The proposed framework therefore provides a technically simple, interpretable, and architecture-compatible approach for reducing temporal instability in exemplar-based video colorization.

**Keywords:** exemplar-based video colorization; temporal consistency; perceptual loss; Bhattacharyya distance; VGG19; bidirectional color fusion; temporal interpolation; deep learning

# 1. Introduction

Colorization aims to assign plausible chromatic information to grayscale visual content. It is an important problem in computer vision and computer graphics because of its applications to historical photograph enhancement, old-film restoration, digital entertainment, cultural-heritage preservation, content creation, and interactive visual editing. Despite significant progress in image colorization, video colorization remains considerably more challenging because a successful method must produce not only visually plausible individual frames but also consistent color trajectories over time.

In single-image colorization, the objective is commonly to infer missing chrominance information from an available luminance image. However, grayscale-to-color reconstruction is fundamentally ill-posed because several chromatic interpretations may be valid for an identical luminance structure. Recent deep colorization methods have therefore incorporated instance-aware reasoning, semantic representations, transformer architectures, and generative color priors to obtain more vivid and semantically meaningful results [1–3, 6]. Instance-aware colorization, for example, separately models object instances to reduce color confusion between foreground objects and surrounding regions [1]. Fast exemplar-based architectures transfer chromatic information by establishing correspondences between a target grayscale image and a color exemplar [2]. Transformer-based methods improve long-range contextual modeling [3], while generative-prior-based approaches can generate vivid colors for structurally complex natural images [6].

Exemplar-based colorization is especially valuable when a user wishes to control the desired appearance through one or more color references. Instead of relying solely on statistical priors learned from training data, the system attempts to identify semantic correspondences between the target and the exemplar and transfers appropriate chromatic information. However, inaccurate correspondence may transfer a color to an incorrect semantic region. This problem becomes more difficult in videos because objects move, disappear, become occluded, change scale, or undergo deformation over time.

Video colorization adds a temporal dimension to the problem. A frame may appear individually realistic while producing an unpleasant video when consecutive frames exhibit inconsistent colors. For instance, the color of clothing may fluctuate slightly from frame to frame, vegetation may alternate between green and yellow tones, or a background may exhibit unstable saturation. These small frame-level variations are perceived as flickering or blinking when displayed sequentially.

Recent video-colorization research has therefore emphasized temporal coherence. VCGAN incorporates recurrent information and long-term loss constraints to balance color vividness and video continuity [4]. Temporally consistent video colorization with deep feature propagation and self-regularization explicitly propagates frame-level deep features bidirectionally and constrains predictions obtained at different temporal steps [5]. These studies demonstrate that temporal coherence cannot be reliably obtained by treating every frame as an independent image.

Other recent methods have investigated key-frame propagation and bidirectional temporal reasoning. Bringing Old Films Back to Life uses a recurrent transformer architecture and adjacent-frame information for temporally coherent restoration and can propagate color from key frames across a video sequence [7]. BiSTNet establishes semantic correspondence between target frames and exemplars and introduces bidirectional temporal feature fusion to improve reference-based video colorization [8]. SVCNet aggregates information from neighboring colorized frames and a long-range initial reference to improve vividness, temporal consistency, and resistance to color bleeding [9].

The NTIRE 2023 Video Colorization Challenge further demonstrated the continuing difficulty of producing simultaneously vivid, accurate, and temporally consistent videos [10]. RTTLC addressed flickering and color-distribution inconsistency through a restored transformer and a test-time local converter [11], while semantic-correspondence-based automatic video colorization used both reference guidance and the immediately preceding colorized frame to improve temporal stability [12].

More recent developments have increasingly focused on long-range dependencies, memory, diffusion models, multimodal guidance, and reference consistency. ColorMNet employs memory-based spatial-temporal feature propagation to establish reliable relationships with distant frames while reducing accumulated propagation errors [13]. OmniFusion combines motion reasoning and diffusion priors for exemplar-based video colorization [14]. LVCD develops a reference-based video diffusion framework for temporally consistent long-sequence line-art colorization [15]. L-C4 introduces language-based video colorization with temporally deformable attention and cross-clip fusion [16]. Palette-guided video diffusion has also been investigated as a mechanism for balancing color richness with long-range temporal coherence [17]. Unified multimodal diffusion-based colorization frameworks further demonstrate the recent movement toward semantic conditioning, generative priors, and cross-frame temporal modeling [18].

Despite this progress, two limitations motivate the present study.

First, perceptual losses used for image and video reconstruction often compare high-dimensional feature maps through an element-wise Euclidean metric. Such a loss is useful for measuring global feature discrepancy but does not explicitly compare feature distributions. In principle, two feature tensors may produce similar aggregate Euclidean errors while exhibiting different activation patterns. For exemplar-based colorization, where semantic correspondence is critical, a distribution-sensitive comparison may provide an additional training signal.

Second, conventional key-frame-based colorization may divide an interval between two reference frames into two regions. One reference dominates the first half and another dominates the second half. At the switching boundary, the reference condition changes discretely. Even when both individual predictions are visually acceptable, this discrete change can produce a visible temporal discontinuity.

To address these limitations, this paper proposes a video colorization framework with two complementary components.

The first is a **distribution-aware perceptual objective**. Deep feature representations extracted from the estimated and ground-truth images are normalized into feature distributions, and their similarity is measured through the Bhattacharyya coefficient and corresponding distance. A temporal feature-consistency loss is simultaneously applied between neighboring generated frames.

The second is a **time-distance weighted bidirectional frame-fusion method**. A target frame between two key frames is colorized with both neighboring references. The predictions are fused using normalized temporal coordinates. A frame close to the left reference receives a large left-reference weight, whereas a frame close to the right reference receives a large right-reference weight. The two weights always sum to one and vary continuously over the interval.

The main contributions are summarized as follows:

1. A distribution-aware spatial perceptual loss is formulated using normalized deep feature activations and Bhattacharyya distance.

2. A temporal perceptual consistency term is introduced to constrain deep-feature variation between adjacent generated frames.

3. A bidirectional frame-colorization mechanism is proposed in which every intermediate frame receives predictions from both neighboring reference frames.

4. A normalized time-distance fusion rule is developed to create a continuous transition of reference influence and eliminate hard reference switching by construction.

5. A result-oriented evaluation framework is presented that clearly distinguishes mathematically guaranteed properties from experimentally testable hypotheses and avoids reporting fabricated numerical values when benchmark measurements are unavailable.

The remainder of this paper is organized as follows. Section 2 reviews related developments in image colorization, exemplar-based colorization, video colorization, temporal consistency, and generative approaches. Section 3 presents the complete proposed framework. Section 4 describes the evaluation methodology and result-oriented analysis. Section 5 discusses limitations and future directions, and Section 6 concludes the paper.

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# 2. Related Work

## 2.1 Deep Image Colorization

Deep image colorization methods estimate missing chrominance values from luminance information using learned visual representations. Modern approaches have evolved from direct pixel-wise regression to semantic, instance-aware, transformer-based, and generative formulations.

Su et al. introduced instance-aware image colorization, recognizing that independent object instances may require separate semantic reasoning to prevent color confusion between adjacent regions [1]. Xu et al. proposed a stylization-based architecture for fast deep exemplar colorization, highlighting the importance of correspondence between a target and its reference exemplar [2]. Kumar et al. introduced the Colorization Transformer, demonstrating the usefulness of transformer-based long-range dependencies for color prediction [3]. BigColor subsequently employed a generative color prior to improve vividness and naturalness in complex images [6].

These works demonstrate two major principles relevant to the present paper. First, semantic information is critical because grayscale intensity alone is insufficient to determine correct colors. Second, high-level feature representations can improve correspondence and color plausibility. However, processing each video frame independently does not guarantee temporal coherence.

## 2.2 Exemplar-Based Colorization

Exemplar-based colorization transfers color information from a provided color reference to a target grayscale image or video. Compared with fully automatic colorization, it offers greater control over the desired output.

The central difficulty is correspondence. A reference and target may contain similar semantic categories but differ in position, viewpoint, illumination, scale, and structure. An inaccurate reference-target correspondence can transfer color to an incorrect region or create color bleeding around object boundaries.

Recent exemplar-based approaches therefore increasingly rely on deep semantic features, attention mechanisms, bidirectional temporal reasoning, and pretrained visual priors. BiSTNet explicitly combines semantic image priors with bidirectional temporal feature fusion [8]. More recent exemplar-based methods, including OmniFusion, further investigate motion modeling and generative diffusion priors [14].

The present method is compatible with such correspondence architectures. Its main contribution does not require replacement of an entire exemplar-colorization backbone; instead, it can augment a backbone with an alternative perceptual objective and a deterministic time-distance fusion stage.

## 2.3 Temporally Consistent Video Colorization

A principal problem in video colorization is flickering. Even small chromatic differences become conspicuous when consecutive frames are rapidly displayed.

VCGAN employs recurrent feedback and dense long-term loss to improve temporal continuity [4]. Liu et al. proposed a unified temporally consistent video colorization framework based on bidirectional deep-feature propagation and self-regularization [5]. SVCNet aggregates short-range neighboring information and long-range initial-frame information [9]. Semantic-correspondence-based methods supervise each frame through both a generated reference and a preceding colorized frame [12].

These methods demonstrate that temporal information can be incorporated through recurrent memory, feature propagation, reference guidance, temporal losses, or combinations thereof. Nevertheless, recurrent propagation may accumulate errors, whereas local cross-frame methods may not fully preserve long-term color identity.

ColorMNet addresses this problem by maintaining a memory that connects spatial-temporal features from distant frames [13]. L-C4 applies temporally deformable attention and cross-clip fusion [16]. Palette-based approaches provide global color guidance to reduce long-range inconsistency [17].

The proposed method takes a complementary direction. Rather than requiring a complex additional recurrent or attention module, it applies explicit bidirectional reference inference and deterministic temporal interpolation.

## 2.4 Bidirectional Temporal Reasoning

Bidirectional processing uses information from both temporal directions. In colorization, it is particularly natural because a frame between two known key frames has access to informative references on both sides.

BiSTNet uses bidirectional temporal feature fusion to improve the use of exemplar colors [8]. Other methods propagate information from adjacent frames, long-term memory, or neighboring clips [5, 9, 13]. These studies motivate the principle that a target frame should not necessarily be determined by only one temporal direction.

The present work differs in that it explicitly computes two independently reference-conditioned chromatic predictions and then combines them using normalized temporal coordinates. This makes the reference influence directly interpretable.

## 2.5 Diffusion-Based Colorization

Recent colorization research has adopted diffusion models because of their strong visual priors and generative capabilities. Reference-based line-art video colorization with diffusion models has demonstrated improved temporal consistency for long animations [15]. L-C4 employs cross-modal representations and temporal mechanisms for creative, semantically controlled video colorization [16]. Palette-guided diffusion methods use global chromatic information to improve both richness and coherence [17], while multimodal systems investigate combinations of grayscale structure, text, palette, and other guidance modalities [18].

These recent approaches provide high generative capacity but can require considerable computational resources. The present framework instead emphasizes a simpler mechanism that can be incorporated into conventional exemplar-based networks and interpreted explicitly.

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# 3. Proposed Method

## 3.1 Problem Formulation

Let a grayscale video sequence be

$$X = \{x_0, x_1, \dots, x_{T-1}\},$$

where $x_t$ is the grayscale frame at temporal index $t$, and $T$ denotes the total number of frames.

The desired colorized sequence is

$$\hat{Y} = \{\hat{y}_0, \hat{y}_1, \dots, \hat{y}_{T-1}\}.$$

Using the CIE Lab color space, the grayscale input provides the luminance channel $L$, whereas the network estimates the chrominance channels $a$ and $b$.

For a target frame $x_t$, an exemplar-based network can generally be expressed as

$$\hat{y}_t = F(x_t, r; \theta),$$

where $r$ denotes a color reference image and $\theta$ represents trainable parameters.

The proposed framework aims to satisfy the following requirements:

1. preserve the spatial structure of the grayscale input;
2. transfer semantically reasonable chromatic information from the reference;
3. reduce color inconsistency between adjacent frames;
4. avoid abrupt transitions when the dominant reference changes;
5. remain compatible with existing exemplar-colorization backbones.

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## 3.2 Framework Overview

The proposed framework contains two principal components.

### Component 1: Advanced perceptual training objective

During training, generated color frames and corresponding ground-truth color frames are processed by a fixed deep feature extractor. The resulting representations are compared through a distribution-aware spatial loss. An additional temporal perceptual term constrains adjacent estimated frames.

### Component 2: Bidirectional time-distance fusion

Between two neighboring key frames, each intermediate target frame is colorized twice:

$$\omega_t^L = F(x_t, r^L; \theta),$$

$$\omega_t^R = F(x_t, r^R; \theta),$$

where $r^L$ and $r^R$ are the left and right color references.

The two outputs are combined using normalized temporal weights.

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## 3.3 Motivation for Distribution-Aware Perceptual Comparison

Conventional perceptual losses generally compare deep features using an $L_1$ or $L_2$ norm. A generic Euclidean feature loss is

$$L_E^{(l)} = \frac{1}{C_l H_l W_l} \left\| \phi_l(y_t) - \phi_l(\hat{y}_t) \right\|_2^2 ,$$

where $\phi_l(\cdot)$ denotes activations of layer $l$, and $C_l$, $H_l$, and $W_l$ denote feature dimensions.

This comparison penalizes position-wise differences. However, it does not explicitly compare the statistical distribution of activations. A distribution-sensitive term can therefore complement the Euclidean formulation.

Let

$$F_l^{gt} = \phi_l(y_t)$$

and

$$F_l^{out} = \phi_l(\hat{y}_t).$$

The tensors are vectorized and transformed into non-negative normalized distributions:

$$P_l(i) = \frac{\exp(F_l^{gt}(i)/\tau)}{\sum_j \exp(F_l^{gt}(j)/\tau)},$$

$$Q_l(i) = \frac{\exp(F_l^{out}(i)/\tau)}{\sum_j \exp(F_l^{out}(j)/\tau)},$$

where $\tau > 0$ is a temperature parameter controlling distribution sharpness.

The Bhattacharyya coefficient is

$$BC(P_l, Q_l) = \sum_i \sqrt{P_l(i) Q_l(i)}.$$

The corresponding Bhattacharyya distance is

$$D_B(P_l, Q_l) = -\ln \left( BC(P_l, Q_l) + \epsilon \right),$$

where $\epsilon$ is a small stability constant.

The proposed spatial perceptual loss is then defined as

$$L_{perc\_spatial} = \sum_{l \in \mathcal{L}} \alpha_l D_B(P_l, Q_l),$$

where $\alpha_l$ specifies the relative importance of layer $l$.

This formulation has an intuitive interpretation:

- if the normalized activation distributions strongly overlap, $BC$ approaches one and the distance approaches zero;
- if the distributions differ, the coefficient decreases and the loss increases.

Therefore, the loss explicitly penalizes distribution mismatch in deep feature space.

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## 3.4 Multi-Layer VGG19 Representation

A fixed VGG19 network can be employed as the feature extractor. Let

$$\mathcal{L} = \{\mathrm{relu1\_2}, \mathrm{relu2\_2}, \mathrm{relu3\_2}, \mathrm{relu4\_2}, \mathrm{relu5\_2}\}.$$

Shallow layers capture local texture, edges, and color boundaries, while deeper layers encode increasingly semantic representations.

Instead of using only a single feature layer, the proposed method aggregates information across several depths:

$$L_{perc\_spatial} = \sum_{l=1}^{L} \alpha_l D_B(P_l, Q_l), \qquad L = 5.$$

The layer weights may be set uniformly initially or determined empirically on a validation set.

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## 3.5 Temporal Perceptual Consistency

Spatial perceptual matching alone cannot guarantee stable colors over time. Therefore, an additional temporal loss is applied between adjacent generated frames.

A direct temporal feature loss is

$$L_{perc\_temporal} = \sum_{l \in \mathcal{L}} \beta_l \frac{1}{C_l H_l W_l} \left\| \phi_l(\hat{y}_t) - \phi_l(\hat{y}_{t-1}) \right\|_2^2 .$$

This formulation is suitable for scenes with relatively small frame-to-frame changes.

For larger motion, a motion-compensated extension is preferable:

$$L^{warp}_{perc\_temporal} = \sum_{l \in \mathcal{L}} \beta_l \left\| M_t^{(l)} \odot \left[ \phi_l(\hat{y}_t) - W\left( \phi_l(\hat{y}_{t-1}), u^{(l)}_{t-1 \to t} \right) \right] \right\|_2^2 ,$$

where:

- $W(\cdot)$ is a warping operation;
- $u_{t-1\to t}$ is estimated optical flow;
- $M_t$ is an optional confidence or visibility mask;
- $\odot$ represents element-wise multiplication.

The non-warped form is computationally simpler, whereas the motion-compensated extension is more appropriate for large displacements.

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## 3.6 Overall Perceptual Objective

The total perceptual loss is

$$L_{perc} = \lambda_s L_{perc\_spatial} + \lambda_t L_{perc\_temporal},$$

where $\lambda_s$ and $\lambda_t$ control spatial and temporal importance.

A complete training objective may further include a chrominance reconstruction term:

$$L_{ab} = \|\hat{a}_t - a_t\|_1 + \left\|\hat{b}_t - b_t\right\|_1 .$$

The overall objective becomes

$$L_{total} = \lambda_{ab} L_{ab} + \lambda_s L_{perc\_spatial} + \lambda_t L_{perc\_temporal} + \lambda_{reg} L_{reg},$$

where $L_{reg}$ may optionally denote smoothness or other architecture-specific regularization.

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## 3.7 Conventional Between-Frame Reference Switching

Consider two neighboring color key frames separated by $N-1$ temporal steps.

A simple conventional strategy selects the left reference for approximately the first half of the interval and the right reference for the second half:

$$r_t = \left\{ r^L, \quad t < m,\ r^R,\ t \geq m, \right.$$

where $m$ is a midpoint.

This introduces a discontinuity in the conditioning reference at $m$. If

$$F(x_m, r^L)\ /\ F(x_m, r^R),=$$

an abrupt chromatic transition may occur at the switching point.

The problem is therefore structural: the reference condition itself changes discontinuously.

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## 3.8 Proposed Bidirectional Colorization

For every frame $x_t$ between two key frames, two color predictions are generated:

$$\omega_t^L = F(x_t, r^L; \theta),$$
$$\omega_t^R = F(x_t, r^R; \theta).$$

The temporal distances to the references are

$$d_t^L = t,$$

and

$$d_t^R = N - 1 - t.$$

The normalized temporal coordinate is

$$\alpha_t = \frac{t}{N-1}.$$

Thus,

$$1 - \alpha_t = \frac{N-1-t}{N-1}.$$

The proposed fused chromatic prediction is

$$p_t = (1-\alpha_t)\omega_t^L + \alpha_t \omega_t^R.$$

Equivalently,

$$p_t = \frac{N-1-t}{N-1}\omega_t^L + \frac{t}{N-1}\omega_t^R.$$

For $t = 0$,

$$p_0 = \omega_0^L.$$

For $t = N - 1$,

$$p_{N-1} = \omega_{N-1}^R.$$

At the midpoint, both references contribute approximately equally.

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## 3.9 Mathematically Demonstrable Properties of the Fusion Rule

The principal advantage of the fusion stage can be demonstrated without requiring experimental measurements.

### Property 1: Convexity

For all $t \in [0, N-1]$,

$$0 \leq \alpha_t \leq 1$$

and

$$(1 - \alpha_t) + \alpha_t = 1.$$

Therefore, $p_t$ is a convex combination of the two reference-conditioned predictions.

### Property 2: Boundary preservation

At the first frame,

$$p_0 = \omega_0^L,$$

and at the final frame,

$$p_{N-1} = \omega_{N-1}^R.$$

Thus, each interval endpoint preserves the prediction associated with its nearest key reference.

### Property 3: Continuous weight transition

The change in the right-reference weight between consecutive frames is

$$\alpha_{t+1} - \alpha_t = \frac{1}{N-1}.$$

Similarly,

$$(1 - \alpha_{t+1}) - (1 - \alpha_t) = -\frac{1}{N-1}.$$

Therefore, the two reference weights vary uniformly rather than switching abruptly.

### Property 4: No hard midpoint reference switch

A conventional half-interval strategy uses a discrete reference-selection operation. The proposed rule does not contain such a discrete switch. Reference influence evolves continuously throughout the interval.

### Property 5: Architectural compatibility

Because fusion is applied to two reference-conditioned outputs, the method does not intrinsically require redesign of the backbone. Any exemplar-based model capable of producing

$$F(x_t, r^L) \quad \text{and} \quad F(x_t, r^R)$$

can theoretically use the proposed fusion rule.

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## 3.10 Algorithm

**Algorithm 1: Time-distance weighted bidirectional video colorization**

**Input:**
Grayscale sequence $X$; key references $R$; exemplar-colorization model $F$.

**Output:**
Colorized video $\hat{Y}$.

1. Select or obtain neighboring reference frames $r^L$ and $r^R$.
2. Determine the interval of $N$ frames.
3. For each frame index $t = 0, \ldots, N-1$:
4. Generate a left-conditioned prediction:

$$\omega_t^L = F(x_t, r^L).$$

5. Generate a right-conditioned prediction:

$$\omega_t^R = F(x_t, r^R).$$

6. Calculate

$$\alpha_t = \frac{t}{N-1}.$$

7. Calculate the fused color map:

$$p_t = (1 - \alpha_t)\omega_t^L + \alpha_t \omega_t^R.$$

8. Combine $p_t$ with the luminance channel.
9. Convert the resulting Lab image to RGB.
10. Repeat for all intervals.
11. Return the complete colorized video.

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# 4. Experimental Evaluation and Result-Oriented Analysis

## 4.1 Evaluation Status and Scientific Reporting Principle

A complete SCI submission should ideally include benchmark results obtained by executing all compared methods under controlled conditions. However, the current development environment does not provide sufficient resources to obtain a complete set of full-scale experimental measurements.

Therefore, this manuscript deliberately does **not** fabricate PSNR, SSIM, LPIPS, temporal-consistency, or runtime values.

Instead, this section provides four forms of technically valid evidence:

1. mathematically demonstrable characteristics of the proposed fusion rule;
2. component-level analytical comparison with the conventional method;
3. explicit qualitative predictions derived directly from algorithmic behavior;
4. a fully specified experimental protocol that can be executed when adequate computational resources become available.

This distinction is important because a predicted advantage should not be presented as an experimentally measured fact.

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## 4.2 Proposed Experimental Protocol

The proposed method should be evaluated on established video benchmarks containing motion, object deformation, camera movement, occlusion, and scene variation.

Recommended evaluation datasets include:

- DAVIS;
- Videvo;
- standard NTIRE video-colorization test data where access conditions permit;
- an additional real-world old-film collection for qualitative testing.

Every original color video should be converted to grayscale for input, while the original color sequence serves as ground truth.

Key frames should be selected at regular intervals of

$$N \in \{5, 10, 20, 30\},$$

allowing analysis of performance under increasingly sparse references.

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## 4.3 Compared Methods

The following comparison groups should be considered.

### Baseline B0: Independent frame colorization

Every frame is processed without temporal constraints.

### Baseline B1: Single-reference exemplar colorization

Each target frame uses one reference only.

### Baseline B2: Hard half-interval switching

The left reference is used before the midpoint and the right reference after the midpoint.

### Variant V1: Distribution-aware spatial perceptual loss

The model uses the Bhattacharyya-based feature-distribution term but no temporal perceptual loss or weighted frame fusion.

### Variant V2: Temporal perceptual consistency

The baseline is augmented only with the temporal perceptual term.

### Variant V3: Time-distance weighted frame fusion

The baseline network is unchanged, but inference uses predictions generated from both neighboring references and the proposed interpolation.

### Full method

The model uses distribution-aware spatial perceptual loss, temporal feature consistency, and bidirectional time-distance fusion.

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## 4.4 Evaluation Metrics

The following metrics are recommended.

### 4.4.1 PSNR

Peak signal-to-noise ratio evaluates pixel-level reconstruction fidelity. Higher values are better.

### 4.4.2 SSIM

Structural similarity measures luminance, contrast, and structural correspondence. Higher values indicate greater structural fidelity.

### 4.4.3 LPIPS

LPIPS assesses perceptual similarity using learned deep features. Lower values indicate stronger perceptual similarity.

### 4.4.4 Color Distribution Consistency

A temporal color-distribution consistency measure can quantify variations of chromatic distributions across temporal intervals. Lower distribution divergence is preferable when scene content remains consistent.

### 4.4.5 Temporal Warping Error

Optical flow can be used to warp one frame toward the next, after which chromatic discrepancy is calculated in visible regions:

$$E_{warp} = \frac{\sum_t \|M_t \odot (\hat{y}_t - W(\hat{y}_{t-1}, u_{t-1\to t}))\|_1}{\sum_t \|M_t\|_1 + \epsilon}.$$

A smaller value indicates stronger motion-compensated temporal consistency.

### 4.4.6 Reference-Boundary Discontinuity

Because the proposed method specifically targets midpoint blinking, an interval-boundary metric should be included:

$$E_{boundary} = \frac{1}{K} \sum_{k=1}^{K} D\left(\hat{y}_{m_k-1}, \hat{y}_{m_k}\right),$$

where $m_k$ denotes a conventional reference-switch boundary and $D$ is a motion-aware chromatic difference measure.

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## 4.5 Analytical Comparison of the Original and Proposed Frame-Fusion Schemes

The original half-interval method can be represented as

$$p_t^{hard} = \Big\{\omega_t^L, \quad t < m,\ \omega_t^R,\ t \geq m.$$

At the switching position,

$$\Delta_{hard} = \left\| \omega_m^R - \omega_{m-1}^L \right\|.$$

This difference may be large because the two frames were generated under different reference conditions.

The proposed method is

$$p_t = (1-\alpha_t)\omega_t^L + \alpha_t \omega_t^R.$$

The change between consecutive fused frames is

$$p_{t+1} - p_t = (1-\alpha_{t+1})(\omega_{t+1}^L - \omega_t^L) + \alpha_{t+1}(\omega_{t+1}^R - \omega_t^R) + \frac{1}{N-1}(\omega_t^R - \omega_t^L).$$

This expression reveals an important advantage. The change in reference influence is limited to

$$\frac{1}{N-1}$$

per frame. There is no one-step change from complete left-reference dependence to complete right-reference dependence.

Therefore, provided that both reference-conditioned prediction sequences are themselves reasonably stable, the proposed interpolation structurally reduces the risk of a sudden transition caused only by reference switching.

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## 4.6 Result-Oriented Qualitative Comparison

Although complete measured benchmark scores are not available, the expected behavior of each variant can be analyzed directly from its mechanism.

| Method | Semantic feature matching | Adjacent-frame constraint | Two-sided reference use | Hard switching | Expected main behavior |
|---|---|---|---|---|---|
| Independent frame method | Limited | No | No | No | High flicker risk |
| Single-reference method | Reference-dependent | Limited | No | Possible at new interval | Stable near reference, possible drift |
| Hard half-interval method | Reference-dependent | Limited | Yes, but separately | Yes | Possible midpoint blinking |

| Method | Semantic feature matching | Adjacent-frame constraint | Two-sided reference use | Hard switching | Expected main behavior |
|---|---|---|---|---|---|
| Distribution-aware loss only | Improved distribution sensitivity | No explicit temporal constraint | Backbone-dependent | Possible | Better semantic discrimination expected |
| Temporal loss only | Conventional | Yes | Backbone-dependent | Possible | Better local frame coherence expected |
| Weighted fusion only | Conventional | Indirectly through interpolation | Yes | No | Smoother reference transition by construction |
| Full proposed framework | Distribution-aware | Yes | Yes | No | Strongest expected balance of semantic and temporal consistency |

The word **expected** is used intentionally where numerical experimental proof is not yet available.

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## 4.7 Advantage of the Advanced Spatial Perceptual Loss

Consider two candidate generated features $Q_1$ and $Q_2$ relative to ground truth $P$. Conventional Euclidean errors may be similar:

$$\|P - Q_1\|_2 \approx \|P - Q_2\|_2.$$

However, the normalized activation distributions can have different overlap:

$$BC(P, Q_1) \ / \ BC(P, Q_2).=$$

The Bhattacharyya component can therefore provide an additional distinction that is absent from the Euclidean magnitude alone.

For exemplar-based colorization, this is potentially beneficial in cases involving:

- similarly shaped objects with different semantic identities;
- foreground-background ambiguity;
- repeated patterns;
- objects with similar grayscale intensity but different expected colors;
- semantic mismatch between target and exemplar regions.

The proposed loss does not guarantee perfect semantic correspondence by itself, but it provides a theoretically justified distribution-sensitive training signal.

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## 4.8 Advantage of Temporal Perceptual Consistency

An independently colorized sequence optimizes each frame primarily for its own appearance. Consequently, two neighboring predictions can both appear individually plausible while being inconsistent with each other.

The temporal term penalizes this behavior in feature space:

$$L_{perc\_temporal} \propto \|\phi(\hat{y}_t) - \phi(\hat{y}_{t-1})\|^2.$$

This can discourage unnecessary deep-feature fluctuations.

The expected advantages are:

1. reduced frame-to-frame semantic color changes;
2. increased stability for persistent objects;
3. reduced flickering in regions undergoing small motion;
4. complementary behavior with bidirectional fusion.

A limitation is that direct adjacent-feature comparison may over-penalize legitimate motion. Therefore, the motion-compensated extension is recommended for scenes with large displacement.

---

## 4.9 Advantage of Time-Distance Weighted Fusion

The strongest analytically demonstrable benefit of the proposed framework occurs in the fusion stage.

In a hard switching method, reference influence changes as

$$1 \to 0$$

for the first reference and

$$0 \to 1$$

for the second reference at a single boundary.

In the proposed method, the left-reference weight follows

$$1,\ 1 - \frac{1}{N-1},\ 1 - \frac{2}{N-1}, \ldots, 0,$$

while the right-reference weight follows

$$0,\ \frac{1}{N-1},\ \frac{2}{N-1},\ldots,1.$$

Thus, influence is transferred progressively across the interval.

This property directly supports the intended reduction of blinking caused specifically by abrupt changes in key-frame conditioning.

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## 4.10 Analysis of Representative Temporal Positions

Consider an interval with $N = 11$ frames.

The normalized right-reference weight is

$$\alpha_t = \frac{t}{10}.$$

The weights are therefore:

| Frame $t$ | Left weight | Right weight |
|---:|---:|---:|
| 0 | 1.0 | 0.0 |
| 1 | 0.9 | 0.1 |
| 2 | 0.8 | 0.2 |
| 3 | 0.7 | 0.3 |
| 4 | 0.6 | 0.4 |
| 5 | 0.5 | 0.5 |
| 6 | 0.4 | 0.6 |
| 7 | 0.3 | 0.7 |
| 8 | 0.2 | 0.8 |
| 9 | 0.1 | 0.9 |
| 10 | 0.0 | 1.0 |

No hard midpoint transition occurs.

This simple numerical example illustrates the core difference from the original method without pretending to be a measured benchmark result.

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## 4.11 Ablation Hypotheses

A future controlled ablation should test the following hypotheses.

### H1: Distribution-aware spatial loss

Adding the Bhattacharyya-based term should primarily improve perceptual or semantic similarity measurements rather than directly guaranteeing temporal coherence.

### H2: Temporal perceptual loss

Adding the temporal term should primarily improve inter-frame consistency metrics, particularly in low- and medium-motion regions.

### H3: Weighted fusion

Replacing hard reference switching with temporal interpolation should reduce discontinuity around interval boundaries.

### H4: Full method

Combining all components should provide the best balance between:

- frame-level visual fidelity;
- semantic color consistency;
- short-term temporal smoothness;
- key-frame interval continuity.

These hypotheses must be tested experimentally before being presented as quantitative findings.

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## 4.12 Recommended Ablation Table for Final Benchmark Execution

| Variant | Bhattacharyya spatial loss | Temporal perceptual loss | Weighted fusion | PSNR ↑ | SSIM ↑ | LPIPS ↓ | Temporal error ↓ | Boundary error ↓ |
|---|---|---|---|---|---|---|---|---|
| Baseline | No | No | No | TBD | TBD | TBD | TBD | TBD |
| A | Yes | No | No | TBD | TBD | TBD | TBD | TBD |
| B | No | Yes | No | TBD | TBD | TBD | TBD | TBD |
| C | No | No | Yes | TBD | TBD | TBD | TBD | TBD |
| D | Yes | Yes | No | TBD | TBD | TBD | TBD | TBD |

| Variant | Bhattacharyya spatial loss | Temporal perceptual loss | Weighted fusion | PSNR ↑ | SSIM ↑ | LPIPS ↓ | Temporal error ↓ | Boundary error ↓ |
|---|---|---|---|---|---|---|---|---|
| Full method | Yes | Yes | Yes | TBD | TBD | TBD | TBD | TBD |

No numerical entry should be filled without actual computation.

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## 4.13 Recommended Visual Evaluation

For every test sequence, the following rows should be presented:

1. grayscale input;
2. left reference;
3. right reference;
4. original hard-switching result;
5. proposed weighted-fusion result;
6. full proposed framework;
7. ground truth.

The columns should contain consecutive frames around a conventional reference-switching boundary.

The strongest cases for demonstrating the method are expected to include:

- clothing with a stable identity;
- vehicles crossing an interval;
- faces and skin tones;
- vegetation;
- moving foreground objects;
- camera pans;
- background regions with similar grayscale intensities but distinct colors.

The most important visual question is not simply whether an individual frame is attractive, but whether the same semantic object preserves stable chromatic identity across time.

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## 4.14 Expected Failure Cases

The proposed method does not solve every video-colorization problem.

### Different object colors in the two references

Suppose the same object is red in $r^L$ and blue in $r^R$. Linear fusion may produce an intermediate purple color that is temporally smooth but semantically incorrect.

### Newly appearing object

An object absent from both key references cannot receive reliable exemplar color information.

### Scene cut

Time-distance fusion should not be applied across a hard scene boundary.

### Large motion

Direct temporal feature comparison without warping can penalize valid motion.

### Inaccurate reference correspondence

If both reference-conditioned predictions contain incorrect colors, averaging cannot recover the correct answer.

### Computational cost

Bidirectional inference nominally requires two reference-conditioned predictions per intermediate frame. Feature caching and parallel batching should therefore be investigated.

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# 5. Discussion, Limitations, and Future Research

The proposed approach combines training-time and inference-time mechanisms.

The advanced perceptual loss addresses feature-space representation. Its purpose is to provide a distribution-sensitive penalty in addition to conventional pointwise discrepancy. The temporal perceptual term addresses consistency between neighboring predictions. The fusion method addresses a different issue: abrupt changes in reference influence.

This division is important because flickering can arise from multiple causes. A poor frame-level colorizer may produce inconsistent semantic assignments. A temporally unaware model may fluctuate even when individual predictions are good. A key-frame pipeline may create a sudden discontinuity simply because its conditioning reference changes abruptly.

The proposed framework addresses all three mechanisms:

$$\text{semantic feature mismatch} \rightarrow L_{perc\_spatial},$$

$$\text{adjacent-frame inconsistency} \rightarrow L_{perc\_temporal},$$

$$\text{hard reference switching} \rightarrow \text{time-distance fusion}.$$

Several improvements remain possible.

First, temporal alignment should be integrated for sequences with significant motion. Optical flow or feature correspondence could warp adjacent representations before temporal comparison.

Second, the linear interpolation rule could be generalized. A learned confidence-aware rule could be expressed as

$$p_t = \frac{c_t^L w_t^L \omega_t^L + c_t^R w_t^R \omega_t^R}{c_t^L w_t^L + c_t^R w_t^R + \epsilon},$$

where $c_t^L$ and $c_t^R$ are correspondence-confidence scores.

Third, scene-change detection should disable cross-boundary fusion when references belong to semantically unrelated scenes.

Fourth, motion-aware nonlinear weights could replace linear interpolation:

$$w_t^L = \frac{\exp(-\gamma d_t^L)}{\exp(-\gamma d_t^L) + \exp(-\gamma d_t^R)},$$

$$w_t^R = 1 - w_t^L.$$

Fifth, future work should compare the lightweight deterministic fusion mechanism against recent memory-based, transformer-based, and diffusion-based approaches.

Finally, complete experimental execution remains essential before journal submission. The present analysis provides a technically explicit and reproducible foundation, but claims of numerical superiority must be supported by actual benchmark measurements.

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# 6. Conclusion

This paper presented a temporally consistent exemplar-based video colorization framework combining a distribution-aware perceptual loss with time-distance weighted bidirectional frame fusion.

The first contribution is an advanced spatial perceptual formulation based on normalized deep feature distributions and Bhattacharyya distance. Unlike a purely element-wise Euclidean feature comparison, the proposed term explicitly measures distributional overlap. A temporal perceptual consistency term is additionally incorporated to discourage unnecessary feature fluctuations between neighboring colorized frames.

The second contribution is a bidirectional reference-fusion mechanism. Each target frame between neighboring key frames is colorized using both references. The resulting predictions are combined through weights determined by normalized temporal distance. These weights form a convex combination, preserve endpoint behavior, and change uniformly across the interval. Consequently, the proposed fusion method eliminates the hard reference switch of a conventional half-interval strategy by construction.

Under the present computational limitations, unmeasured quantitative scores have intentionally not been fabricated. Instead, the paper identifies mathematically guaranteed properties, derives the temporal behavior of the fusion weights, presents a reproducible evaluation protocol, defines appropriate quantitative metrics, and states experimentally testable hypotheses.

The analysis indicates that the most direct advantage of the proposed frame-fusion strategy is its continuous transfer of reference influence across time. The perceptual-loss components are designed to complement this property by improving semantic feature comparison and neighboring-frame consistency.

Future work should perform full benchmark evaluation, incorporate motion-compensated temporal alignment, investigate confidence-adaptive fusion, detect scene changes automatically, and compare the proposed lightweight mechanism against recent transformer-, memory-, and diffusion-based video-colorization frameworks.

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